

1. [5 puntos] Considere el sistema de ecuaciones

$$\begin{cases} x + 2y + 3z = 7 \\ x + z = y \\ x + 5y + 5z - 14 = 0 \end{cases}$$

donde x, y, z son incógnitas. Utilice el método de Gauss-Jordan

$$\begin{cases} x + 2y + 3z = 7 \\ x - y + z = 0 \\ x + 5y + 5z = 14 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 1 & -1 & 1 & 0 \\ 1 & 5 & 5 & 14 \end{array} \right)$$

$$\begin{array}{l} -F1 + \widetilde{F2} \\ -F1 + \widetilde{F3} \end{array} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & -3 & -2 & -7 \\ 0 & 3 & 2 & 7 \end{array} \right)$$

$$\begin{array}{l} -2 \cdot F2 + \widetilde{F1} \\ -\frac{1}{3} \cdot \widetilde{F2} \\ -3 \cdot F2 + \widetilde{F3} \end{array} \left(\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & \frac{7}{3} \\ 0 & 1 & \frac{2}{3} & \frac{7}{3} \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\begin{cases} x + \frac{5}{3}z = \frac{7}{3} \rightarrow x = \frac{7}{3} - \frac{5}{3}z \\ y + \frac{2}{3}z = \frac{7}{3} \rightarrow y = \frac{7}{3} - \frac{2}{3}z \end{cases}$$

$$S = \left\{ \left(\frac{7}{3} - \frac{5}{3}z, \frac{7}{3} - \frac{2}{3}z, z \right), z \in \mathbb{R} \right\}$$

2. [4 puntos] Considere las matrices $A = \begin{pmatrix} 3 & 0 & 2 \\ 2 & -1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 2 & -3 \\ -4 & 0 \\ 2 & -1 \end{pmatrix}$ y $D = \begin{pmatrix} 6 & 1 \\ 3 & 1 \\ 3 & 0 \end{pmatrix}$.

Determine la matriz B tal que $A^T \cdot B - C = D$.

$$A = \begin{matrix} F & C \\ 2 & 3 \end{matrix}, A^T = \begin{matrix} F & C \\ 3 & 2 \end{matrix} \rightarrow B = \begin{matrix} F & C \\ 2 & 2 \end{matrix}$$

$$B = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

$$2a+c=5 \rightarrow 2a+1=5 \rightarrow 2a=4 \rightarrow a=2$$

$$2b+d=-1 \rightarrow 2b+(-1)=-1 \rightarrow 2b=0 \rightarrow b=0$$

$$a=2, b=0, c=1, d=-1$$

5. [5 puntos] Determine $z \in \mathbb{C}$ que satisfice simultáneamente las siguientes condiciones :

$$\blacksquare |\bar{z} + (1-i)| = 5$$

$$\blacksquare \text{Arg}(z - (1+2i)) = \frac{3\pi}{4}$$

$$z = a+bi$$

$$\bar{z} = a-bi$$

$$|a-bi+1-i| = 5$$

$$|(a+1)+(-b-1)i| = 5$$

$$(a+1)^2 + (-b-1)^2 = 25, \text{ (entire)}$$

$$\text{Arg}(z - (1+2i)) = \frac{3\pi}{4} \rightarrow 135^\circ$$

$$a+bi-1-2i = \frac{3\pi}{4}$$

$$90^\circ < 135^\circ < 180^\circ$$

$$(a-1)+(b-2)i = \frac{3\pi}{4}$$

$$\pi \text{ (cuadrante)}$$

$$\frac{b-2}{a-1} = \tan\left(\frac{3\pi}{4}\right)$$

$$a < 0 \quad b > 0$$

$$a-1$$

$$\frac{b-2}{a-1} = -1$$

$$a-1$$

$$b-2 = 1-a$$

$$\boxed{b = -1-a}$$

$$1-a = b-2$$

$$-a = b-3$$

$$\boxed{a = 3-b}$$

$$(a+1)^2 + (-b-1)^2 = 25$$

$$(3-b+1)^2 + (-b-1)^2 = 25$$

$$(4-b)^2 + (-b-1)^2 = 25$$

$$16 - 8b + b^2 + b^2 + 2b + 1 = 25$$

$$2b^2 - 6b - 8 = 0$$

$$2b^2 - 6b + 7 = 25$$

$$2b^2 - 6b - 8 = 0$$

$$\begin{array}{r} 2b \\ b \end{array} \begin{array}{r} \times 2 = 2b \\ \times -4 = -8b \end{array}$$

$$-6b$$

$$(2b+2)(b-4)=0$$

$$b=-1 \quad b=4 \quad b>0$$

$$a = b - 3$$

$$= 4 - 3 = 1 \quad a < 0$$

$$a = -1$$

$$z = -1 + 4i$$

6. [4 puntos] Factorice en $\mathbb{C}$ el polinomio $P(z) = z^4 - 90z + 35z^2 - 10z^3 + 234$ si se sabe que $z = 5 - i$ es un cero de $P(z)$.

$$\begin{array}{r|rrrrr} z^4 & -10z^3 & +35z^2 & -90z & +234 & \\ 1 & -10 & 35 & -90 & 234 & \\ \hline & 5-i & -26 & 95-9i & -234 & 5-i \\ \hline 1 & -5-i & 9 & -95-9i & 0 & (x-(5-i)) \end{array}$$

$$(5-i)(-5-i) = -25 - 5i + 5i + i^2 = -26$$

$$\begin{array}{r|rrrrr} x^3 & +(-5-i)x^2 & +9x & -95-9i & & \\ 1 & -5-i & 9 & -95-9i & & (-95-9i)(-5-i) \\ \hline & 5+i & 0 & 95+9i & & 5+i \\ \hline 1 & 0 & 9 & 0 & & (x-(5+i)) \end{array}$$

$$(-95-9i)(-5-i) = -225 + 95i - 45i + 9i^2 = -234$$

$$x^2 + 9 = 0$$

$$(x-3i)(x+3i) = 0$$

$$x = 3i \quad x = -3i$$

$$(x-(5-i))(x-(5+i))(x-3i)(x+3i)$$

$$\boxed{\mathbb{R} / (x-5+i)(x-5-i)(x-3i)(x+3i)}$$

7. [5 puntos] Calcule y exprese el número complejo $z = (1+i)^{-i} + (2\sqrt{3} - 2i)^5$ en la forma rectangular.

$$z = (1+i)^{-i} + z = 2\sqrt{3} - 2i$$

$$\ln(z) = (-i) \ln(1+i) \quad a = 2\sqrt{3} \\ b = -2$$

$$z = 1+i \quad r = \sqrt{(2\sqrt{3})^2 + (-2)^2} = 4$$

$$a = 1 \quad \theta = \arctan\left(\frac{-2}{2\sqrt{3}}\right) = -\frac{\pi}{6} \\ b = -1$$

$$r = \sqrt{1^2 + 1^2} = \sqrt{2} \quad z = \left(4 \left(i\right) \left(-\frac{\pi}{6}\right)\right)^5$$

$$\theta = \arctan(1) = \frac{\pi}{4} \quad = 1024 \left(i\right) \left(-\frac{5\pi}{6}\right)$$

$$z = \sqrt{2} \cdot e^{i\frac{\pi}{4}} \quad 1024 \cos\left(-\frac{5\pi}{6}\right) + 1024 \sin\left(-\frac{5\pi}{6}\right) i$$

$$-886,87 - 572i$$

$$\ln(z) = (-i) \ln(\sqrt{2} \cdot e^{i\frac{\pi}{4}})$$

$$= (-i) \left[\ln(\sqrt{2}) + i\frac{\pi}{4} \right]$$

$$= -\ln(\sqrt{2}) i - i^2 \frac{\pi}{4}$$

$$= -\ln(\sqrt{2}) i + \frac{\pi}{4}$$

$$e^{\frac{\pi}{4}} \cdot \left(i\right) \left(-\ln(\sqrt{2})\right)$$

$$e^{\frac{\pi}{4}} \cdot \left[\cos(-\ln(\sqrt{2})) + \sin(-\ln(\sqrt{2})) i \right]$$

$$2,06 - 0,78i$$

$$2,06 - 0,78i - 886,87 - 572i$$

$$\boxed{-884,81 - 572,78i}$$